

Detecting Neurocognitive Profiles using Classification

4A13 Project Final Report

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Abstract—Educational prediction models focus on regression learning tasks aimed at forecasting final grades. This approach overlooks behavioural factors associated with academic performance, such as high-functioning learning disabilities and related cognitive burdens. Prediction models focused on performance metrics cannot account for diverse neurocognitive profiles and learning pathways of twice-exceptional or ADHD-diagnosed students. This work pivots from regression to multi-class classification, using behavioural and demographic metadata from the UCI Student Performance Dataset to infer neurodiverse profiles. Feature engineering introduced synthetic labels approximating common neurodiversity markers. A Random Forest model with balanced class weights achieved 90% accuracy with high recall for at-risk neurodiverse groups. Behavioural features do provide sufficient data to infer and identify hidden cognitive profiles to support unmet learning needs.

Keywords—*Classification Learning Task, Random Forest, Feature Engineering, Neurodiversity, Educational Triage*

I. INTRODUCTION

Educational analytics using machine learning (ML) traditionally focus on predicting academic outcomes as a highly performance-centric approach to learning [8]. These ML pipelines forecast final grades using behavioural and demographic features. This is done by supervised regression tasks that model the relationship between the behavioural features and final grades as the target [1]. Models forecasting achievements often neglect the cognitive conditions underlying a student's final grades. Consequently, students with high academic output are misrepresented, as high achievement masks the underlying strain or hidden learning needs.

Educational institutions and systems have persistently been unable to detect neurocognitive diversity and are unfit to tailor learning conditions required to facilitate students with diverse cognitive abilities. For example, a student with twice exceptionality may have high grades; however, their struggle with executive functioning may be unseen or masked [9]. Similarly, a student with high-functioning anxiety may maintain great performance through unsustainable effort. Through regression analysis, these students are considered high

performers, yet the analytic results don't provide mechanisms to identify any accommodations or support needs required.

This paper discards academic performance prediction and pivots towards utilizing behavioural metadata such as social activity, absences, health ratings, and study time to classify students into neurocognitive support profiles. The overarching objective is to provide early triage by identifying anxiety patterns, executive dysfunction, or elevated cognitive burdens that compromise a student's ability to learn effectively. Classification of diverse neurocognitive profiles can design different instructional learning strategies for those with unmet support needs.

II. METHODOLOGY

A. Dataset

The UCI Student Performance Dataset has 395 samples with thirty demographic and behavioural features, and three academic attributes denoting two course grades (G1, G2) and final grades (G3) [1]. The three grade attributes were excluded from the input vector to prevent the model from predicting synthetic cognitive profiles from the performance metadata among other features.

B. Feature Engineering

Public educational datasets do not include clinical labels. To approximate support-relevant patterns, a new target variable (Cognitive_Profile) was defined using synthesized clinical rules aligned with known mental health indicators:

- Attention Deficit Hyperactivity Disorder (ADHD) Markers: 'goout' > 3 or 'absences' > 1
- Generalized Anxiety Disorder (GAD): 'studytime' > 1
- Twice Exceptional (2e): 'G3' > 11
- Learning Disability (autism spectrum disorder (ASD)): 'failures' > 0
- Neurotypical: none of the above conditions met

These rules isolate high effort, high performance, and executive dysfunction, which are markers strongly associated with diverse learning needs.

C. Class Imbalance Correction

Original labelling produced a strong majority-class bias towards the neurotypical profile. Profiles were defined using a strict clinical threshold, but this resulted in a dataset where 87% of samples were neurotypical (345 out of 395).

Cognitive_Profile_Original				
Neurotypical	345			
Learning_Disability	29			
ADHD	13			
Twice_Exceptional_2e	4			
Anxiety_GAD	4			
Name: count, dtype: int64				
Random Forest Classification Report				
	precision	recall	f1-score	support
ADHD	0.00	0.00	0.00	2
Anxiety_GAD	0.00	0.00	0.00	2
Learning_Disability	0.88	0.88	0.88	8
Neurotypical	0.94	1.00	0.97	67
accuracy			0.94	79
macro avg	0.45	0.47	0.46	79
weighted avg	0.89	0.94	0.91	79

Figure 1 Preliminary Strict Clinical Profiles and Report

The model trained on this distribution achieved high accuracy of 94%, but accuracy detracts from the main objective because it failed to detect any minority or “at-risk” (ADHD or GAD) neurodiverse profiles. To correct this, profile rule criteria were expanded to increase minority representation, forcing the dataset to have better distribution. Random Forest was then trained using balanced class weights to penalize misclassifying “at-risk” profiles. Though balanced class weights reduced the overall accuracy to 90%, it successfully met the pivotal objective to correctly identify an “at-risk” minority group.

D. Preprocessing

The synthesized clinical dataset was cleaned using standard Pandas techniques [5]. Categorical features such as ‘reason’ or ‘Mjob’ were cleaned using One-Hot Encoding, and all numerical features were normalized using StandardScaler (Z-score normalization). Normalization ensures features with large ranges like ‘absences’ did not influence the model disproportionately. Using `train_test_split`, the data was split into 80% training and 20% testing sets.

III. MODEL EVALUATION

A. Model Selection

Random Forest Classifier was selected as the model. Traditionally, linear models like logistic regression [2] are common baselines for education analytics, and predictions as linear decision boundaries are assumed. However, human behaviour is inherently nonlinear. For example, high absences can correlate with failure in some students but also correlate with boredom in gifted students. Random Forest is composed of multiple decision trees. Random Forest uses bagging, an ensemble learning method that trains models in parallel using random subsets of data [6]. Generalization is improved as the

algorithm checks out a random subset of features at each split [6]. Trees use majority voting for classification [6]. This structure allows the model to capture nonlinear interactions, making it suitable for small tabular datasets and imbalanced class distributions. Deep Learning Neural Networks and Multilayer Perceptron (MLP) were rejected due to the small sample size from selected dataset (N = 395) as anything where $N < 1000$ poses a significant risk of overfitting [3].

B. Optimization Strategy

To address class imbalance corrections described in Section II. C., Random Forest was instantiated with a parameter: “class_weight=‘balanced’”. Initial experiments showed that the model defaulted almost entirely to the neurotypical majority. This behaviour is expected when strict clinical thresholds generate disproportionately small minority classes. Because the main objective of this project was to evaluate whether behavioural features contain enough markers to a model to distinguish neurodiverse profiles, the dataset was intentionally engineered to boost minority classes rather than preserve the real-world prevalent statistics [9]. Contextually, the large neurotypical subset did cause the classifier to ignore the neurodiverse subsets. Applying balanced class weights counteracts this, so the model is forced to learn decision boundaries for minority groups. The goal was to test whether behavioural features could separate neurodiverse profiles from neurotypical ones.

IV. RESULTS AND DISCUSSION

A. Classification Performance

The Random Forest balanced classifier achieved 90% overall accuracy. Anxiety_GAD showed strong recall with 1.00 precision; ADHD had a 0.90 F1-score, and Twice Exceptional had a 0.93 F1-score. This indicates that with feature engineering, the demographic and behavioural metadata contains enough discriminative structure, which is relevant to cognitive-based support and learning needs.

Cognitive_Profile				
ADHD	160			
Twice_Exceptional_2e	115			
Anxiety_GAD	92			
Neurotypical	19			
Learning_Disability	9			
Name: count, dtype: int64				
Random Forest Classification Report				
	precision	recall	f1-score	support
ADHD	0.81	1.00	0.90	30
Anxiety_GAD	1.00	1.00	1.00	16
Learning_Disability	0.00	0.00	0.00	1
Neurotypical	0.00	0.00	0.00	4
Twice_Exceptional_2e	0.96	0.89	0.93	28
accuracy			0.90	79
macro avg	0.55	0.58	0.56	79
weighted avg	0.85	0.90	0.87	79

Figure 2 Balanced Profiles and Random Forest Report

B. Confusion Matrix

The confusion matrix depicts the prediction errors for the

Balanced model. For the engineered risk groups there is mainly strong correct classifications. The confusion lies with three twice exceptional students and two neurotypical students labelled as ADHD. The model may struggle to identify overlapping behavioural expressions like attention deficit and restlessness. The model indicates a trade-off as shown by the precision of 0.00 for neurotypical profiles. Cost-Sensitive Learning strategy was employed. The model was deliberately designed to be hypersensitive to risk by weighing typical minority classes heavily. Thus, it was expected behaviour to flag some neurotypical students as ADHD which indicates a false positive. In educational triage system where ‘at-risk’ markers for minority classes (neurodiverse profiles) are critical, specificity can be sacrificed for sensitivity.

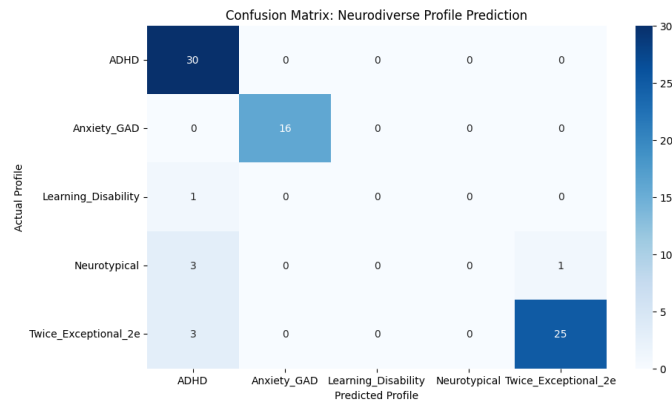


Figure 3 Confusion Matrix

C. Inference Demonstration

To validate a potential minimum viable product and the classifier’s validity an inference simulation was run on sample synthesized student ID 104. This student had moderate academic performance, yet elevated study time and thus was classified under the Anxiety_GAD profile. The model correctly recommended a typical learning strategy for that type of learner. This ideally demonstrates that the model can translate data into immediate actionable support. If scaled up and advanced in education technology firms AI can drive a lot of innovative changes.

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Inference Demonstration
Student ID: 104
AI Detected Profile: Anxiety_GAD
Recommended Intervention:
  Strategy: Low-Stakes
  Recommendation: Untimed Exams to reduce cortisol
  
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Figure 4 Single-Sample Inference

V. CONCLUSION

This paper corroborates that traditional education prediction studies overlook meaningful variation in how students learn. Pivoting from regression to classification validates that it is possible to detect neurocognitive patterning and profiles that indicate unmet learning needs. Through the use of balanced synthetic labels and the Random Forest classifier, the objective was achieved. There was reliable detection of neurodiverse profiles such as GAD, ADHD, and 2e. Though the dataset was synthesized, the results prove that metadata of behaviour and demographics can support proactive identification of students whose hidden traits and learning needs diverge from traditional classroom expectations despite adequate academic outcomes. This AI-driven approach can be scaled up for early learning triage and support. By designing curriculum and learning paradigms that consider cognitive diversity over performance metrics and outcomes, education systems can advance.

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